Deep Learning Image Classification on CIFAR-100

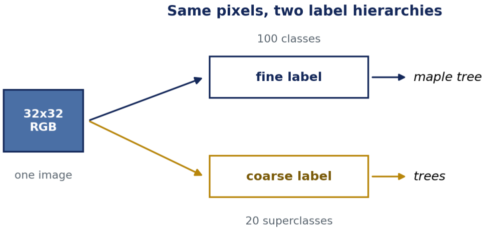
Sequence baselines, CNN controls, transfer learning, and HF ViT

Faidon Grammatikopoulos Konstantinos Dimopoulos Theodoros Kokkas

MSc in Artificial Intelligence
NCSR “Demokritos” & University of Piraeus
Deep Learning Course

June 2026

CIFAR-100: one dataset, two label hierarchies



Standard 50k train / 10k test split, `uoft-cs/cifar100`.

- 60000 natural images, only **32×32** RGB — very little signal per sample.
- Every image carries a **fine** label (100 classes) nested inside a **coarse** superclass (20 classes).
- We study three task styles on the *same* pixels: *binary* one-vs-rest, *coarse* 20-class, *fine* 100-class.

Research question

How do different **architecture families** compare on CIFAR-100 under **limited data** and a **32×32** input, and how far does **transfer learning** move beyond from-scratch baselines?

- A *comparison* study, not a single-model study: sequence models, compact and stronger CNN controls, pretrained CNN families, and a transformer notebook.
- Held constant across families: the data splits, the evaluation protocol, and **Macro F1** as the primary metric.
- What varies: backbone capacity, and how much of it we unfreeze.

Experimental setup

Aspect	Choice
Data	CIFAR-100, 50k train / 10k test, 32×32 RGB
Validation	10% split off the training set
Tasks	binary (1-vs-rest) · coarse (20) · fine (100)
Primary metric	**Macro F1** (+ Top-5 accuracy for multiclass)
Transfer regimes	frozen head $\rightarrow$ partial unfreeze $\rightarrow$ full fine-tune
Optimiser	Adam; light augmentation where configured
Input handling	Scratch at 32 px; EfficientNet at 96–160 px; other transfer / ViT at 224 px
Compute	Google Colab T4 GPU; fixed seed 42

- Macro F1 weights every class equally, so it stays honest on the *balanced* 20- and 100-class tasks *and* on the severely imbalanced 1-vs-rest tasks, where high accuracy can hide poor positive-class recall.

Architecture families surveyed

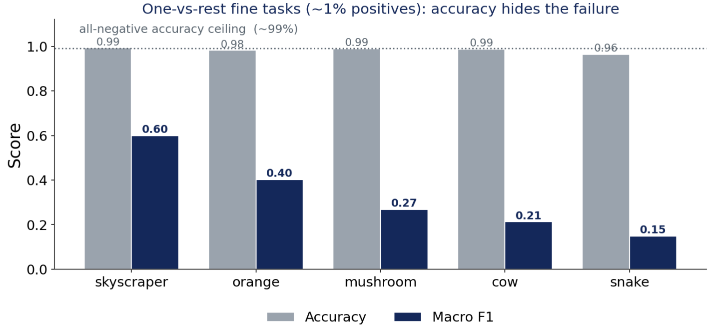
RNN / LSTM Bi-LSTM	Baseline CNN Strong CNN	ResNet50V2 DenseNet121	EfficientNet MobileNetV3	ViT-B/16 (HF)
row sequence	from scratch	classic CNN	transfer CNNs	LoRA
from scratch	*image controls*	*pretrained*	*pretrained*	*standalone*

- The local code compares **row-sequence models**, from-scratch CNN controls, and several pretrained image backbones.
- Pretrained backbones are evaluated under frozen, partially unfrozen, or fine-tuned regimes when the corresponding runs exist.

Narrative arc

From-scratch controls → transfer lift → transformer transfer.

Warm-up: why accuracy misleads under imbalance



- A fine class is $\sim$**1%** of the data, so predicting "always negative" already scores $\sim$**99%** accuracy.
- The baseline CNN sits near that ceiling on every task, yet Macro F1 ranges from **0.60** (skyscraper) down to **0.15** (snake).
- Accuracy hides the failure; **Macro F1** exposes it. This is why it is our primary metric everywhere that follows.

Best baseline-CNN run per one-vs-rest fine task (results//metrics.json).*

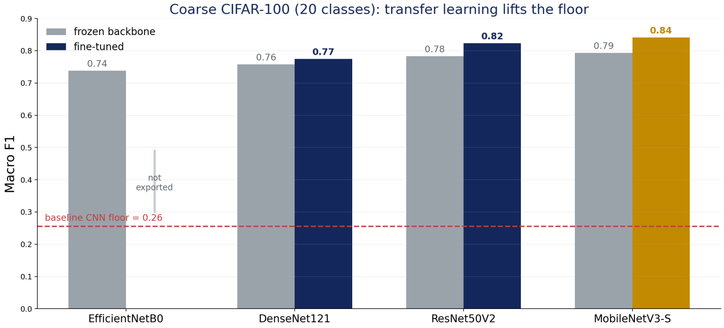
From-scratch controls on 20 coarse classes

Run	Input view	Accuracy	Macro F1	Top-5
Baseline CNN	image $(32, 32, 3)$	0.281	0.263	0.662
Strong CNN, regularized + aug.	image $(32, 32, 3)$	0.283	0.274	0.656
LSTM sequence	rows $(32, 96)$	0.329	0.311	0.720
Bi-LSTM sequence	rows $(32, 96)$	**0.356**	**0.345**	**0.741**

- The sequence models are useful **from-scratch controls**: they read each image as 32 row-timesteps with 96 features per row.
- The stronger CNN improved over the compact baseline, but did not close the gap to LSTM/Bi-LSTM in this recorded coarse run.
- This is a project-specific baseline finding, not a claim that recurrent models are generally stronger image classifiers than CNNs.

Local results: baseline CNN, strong CNN, LSTM, and Bi-LSTM coarse multiclass metrics.

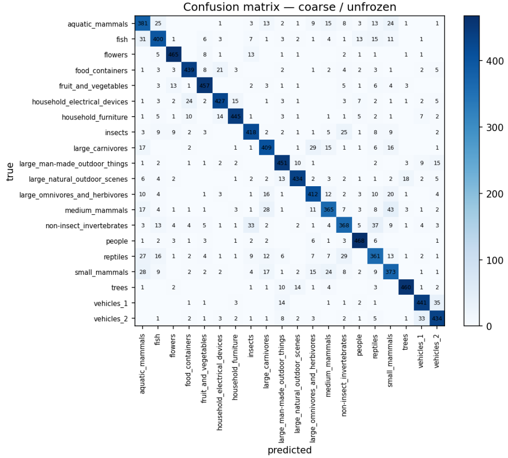
Coarse: transfer learning lifts the floor



- Every pretrained backbone clears the **0.26** baseline by a wide margin, landing in the **0.74–0.84** Macro F1 band.
- Unfreezing helps consistently: fine-tuning beats the frozen head for every family where both regimes were run.

ResNet50V2 / DenseNet121 / EfficientNetB0 / MobileNetV3-Small, coarse multiclass. EfficientNetB0 coarse fine-tune was not exported.

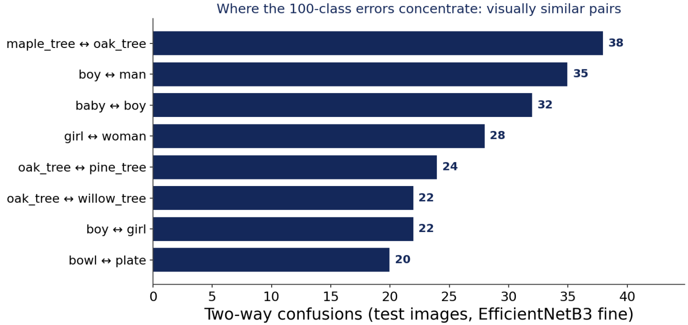
Coarse winner: MobileNetV3-Small, fully unfrozen



- **Macro F1 = 0.8409**, Top-5 **0.983** — the best coarse result of the study.
- The lightest *pretrained* backbone (2.5 M params) wins coarse once fully unfrozen (+0.047 over its frozen head).
- A strong, mostly-diagonal confusion matrix; residual errors fall between semantically close superclasses.

nb06 MobileNetV3-Small, coarse, unfrozen (results/mobilenetv3_coarse/).

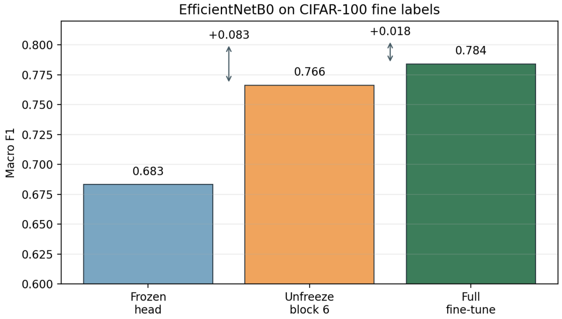
Fine task: why 100 classes is harder



Two-way confusions from the EfficientNetB3 fine confusion matrix.

- **5$\times$ more classes** on the same 32$\times$32 input: the decision boundary gets far finer.
- Errors are not random — they **concentrate within superclasses**: *maple*$\leftrightarrow$*oak*, *boy*$\leftrightarrow$*man*, *girl*$\leftrightarrow$*woman*.
- So the fine problem is really about separating *visually similar* members of the same coarse group.

EfficientNetB0 fine: how far unfreezing goes

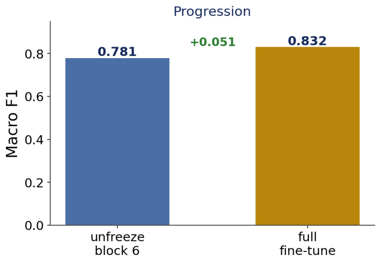


- Frozen head already reaches **0.683**; unfreezing block 6 adds **+0.083**, full fine-tune a further **+0.018**.
- **+0.101 Macro F1** end-to-end — the largest frozen→tuned lift of any family, and the template for the B3 run next.

nb05 EfficientNetB0 fine, input 128 (frozen / unfreeze block6 / full ft).

Best fine ConvNet: EfficientNetB3, full fine-tune

EfficientNetB3, full fine-tune (input 160, 40 ep)



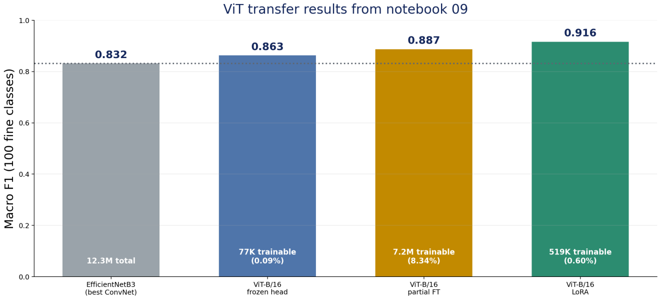
Best fine-class ConvNet

Accuracy	**0.8321**
Macro F1	**0.8319**
Top-3 acc.	**0.9524**
Top-5 acc.	**0.9738**

- **Macro F1 = 0.8319**, Top-5 **0.974** — the strongest *convolutional* fine result.
- A larger pretrained backbone (12.3 M) plus a larger input (160 px) closes much of the 100-class gap.
- Clear winner among the TensorFlow/Keras ConvNet fine multiclass runs.

nb08 EfficientNetB3 fine, full fine-tune, input 160, 40 epochs.

HF ViT-B/16: transformer transfer workflow



nb09 Hugging Face ViT notebook, CIFAR-100 fine test split.

Notebook 09 results

- Frozen head: **0.8629** Macro F1.
- Partial fine-tune: **0.8873** Macro F1.
- LoRA: **0.9165** Macro F1, Top-5 **0.9919**.
- Different stack: PyTorch, Transformers, HF processor at 224 px.

Transformer caveats: how to compare ViT fairly

Best fine result

ViT-B/16 + LoRA

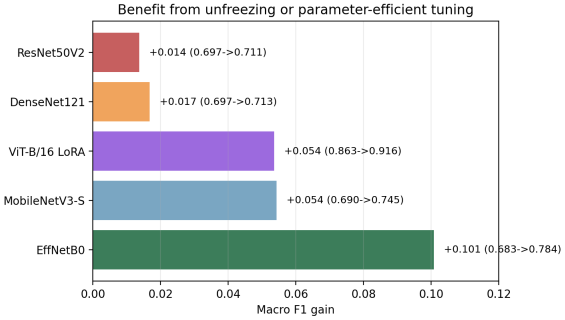
Macro F1 = 0.9165

Top-5 = 0.9919

- ViT changes more than the architecture: the processor resizes CIFAR-100 from 32×32 to 224×224 and applies transformer-specific normalization.
- The strongest ConvNet result remains **EfficientNetB3** with Macro F1 **0.8319**; ViT moves the ceiling higher.
- The comparison is meaningful on the same CIFAR-100 test labels, but the framework, input size, and pretraining prior are different.

nb09 ViT results; nb08 EfficientNetB3 result for the ConvNet reference.

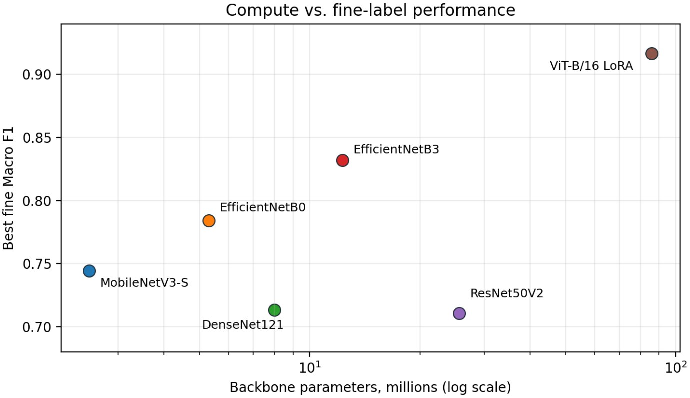
When does unfreezing or PEFT pay off?



- Unfreezing helps the fine task for the families where frozen and tuned results were both exported.
- Largest ConvNet gain: **EfficientNetB0** (0.683 $\rightarrow$ 0.784); MobileNetV3-S and ViT-LoRA both add about **+0.054**.

Per-architecture frozen/head-only $\rightarrow$ best tuned Macro F1 (fine task). ViT uses LoRA, not layer unfreezing.

Compute vs. accuracy trade-off



- Best fine Macro F1 against backbone size reveals a clear **Pareto frontier**.
- **MobileNetV3-S** is the value pick (2.5 M, 0.74); **EfficientNetB3** the sweet spot (12.3 M, 0.83).
- **ViT-B/16 LoRA** is the fine-task leader (Macro F1 0.917), but it also changes framework, input resolution, and pretraining scale.

Parameters: standard published backbone sizes; F1 from local metrics and nb09.

Conclusions: one table, every architecture

Architecture	Binary F1 (1-vs-rest)	Coarse F1 (20 cls)	Fine F1 (100 cls)	Family
Baseline CNN	0.60	0.26	—	scratch
Strong CNN	—	0.274	—	scratch
Bi-LSTM	—	0.345	—	scratch
MobileNetV3-S	—	**0.841**	0.745	transfer
EfficientNetB0	—	0.739	0.784	transfer
DenseNet121	0.77	0.775	0.713	transfer
EfficientNetB3	—	—	0.832	transfer
ResNet50V2	0.91	0.824	0.711	transfer
ViT-B/16 LoRA	—	—	**0.917**	transformer

- **Coarse winner:** MobileNetV3-S (0.841) — smallest *pretrained* backbone wins coarse.
- **Fine winner:** ViT-B/16 LoRA (0.917); best ConvNet is EfficientNetB3 (0.832).
- Fine-tuning usually improves the exported transfer runs; capacity matters most on fine labels.
- Bigger is not automatically better; ResNet50V2 is large yet mid-pack on fine.

Best Macro F1 per task from exported metrics; binary column = most separable 1-vs-rest class.

Future work

Modeling extensions

- **Stronger sequence studies** — tune row-wise LSTM/Bi-LSTM and compare against patch-sequence variants.
- **ViT from scratch** — quantify exactly how much of the transformer's lead is ImageNet pretraining rather than architecture.
- **Parameter-efficient tuning** — repeat the LoRA setting with multiple seeds and other transformer backbones.

Sharper evaluation

- **Augmentation study** — a controlled grid (none $\rightarrow$ light $\rightarrow$ strong) instead of a single fixed recipe.
- **Per-class thresholds** — tune decision thresholds for the imbalanced 1-vs-rest tasks rather than a flat 0.5.
- **Multi-seed runs** — error bars on the headline numbers and on any ViT and transfer-CNN deltas.

One-line takeaway

Transfer learning makes CIFAR-100 tractable; the open questions are *efficiency* and *robustness*, not raw accuracy.

Thank you

Questions & Discussion

Architectural Comparison on CIFAR-100

Binary · Coarse (20) · Fine (100)
CNN and Transformer families

Code & full results:

`github.com/Fgram-devAI/deepl-cifar100-image-analysis`

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